CS & IT ENGINEERING

Operating System

CPU Scheduling

Lecture -5

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Recap of Previous Lecture

Topic Multilevel Queue Scheduling

Topic Multilevel Feedback Queue Scheduling

Topics to be Covered

Topic Priority based Scheduling

Topic Round Robin Scheduling

Topic : Round Robin (RR)

$Q = 3$

Process	Arrival Time	Burst Time
P1	0	12
P2	0	5
P3	3	9
P4	5	6
P5	2	8
P6	4	2
P7	1	7

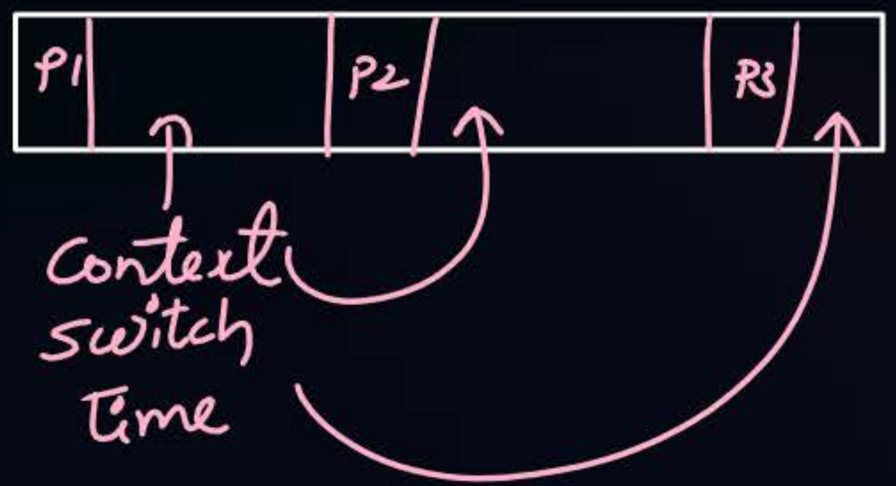
Time	R. Q.
0	~~P1~~, P2
3	~~P2~~, P7, P5, P3, P1
6	P7, P5, P3, P1, P6, P4, P2

P1	P2	P7	P5	P3	P1	P6	P4	P2	P7	P5	P3	P1	P4	P7	P5	P3	P1

0 3 6 9 12 15 18 20 23 25 28 31 34 37 40 41 43 46 49

Topic : What Should Be the Quantum Value?

very-very small



CPU efficiency $\simeq 0$

small

$\Downarrow$

Interactive

large

$\Downarrow$

less interactive

very-very large

RR degrades to FCFS

Topic : RR

Advantages:

1. All processes execute one by one, so no starvation
2. Better interactiveness
3. Burst time is not required to be known in advance

Disadvantages:

1. Average waiting time and turnaround time ~~is~~ are more
2. Can degrade to FCFS

$$\text{CPU efficiency} = \frac{q}{q+s}$$

q = quantum
s = context switch time

[MCQ]

11. (a)

#Q. A scheduling algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (the lowest priority). The scheduler re-evaluates the process priorities every T time units and decides the next process to schedule. Which one of the following is TRUE if the processes have no I/O operations and all arrive at time zero?

GATE-2013

A. This algorithm is equivalent to the first-come-first-serve algorithm

B. This algorithm is equivalent to the round-robin algorithm.

C. This algorithm is equivalent to the shortest-job-first algorithm

D. This algorithm is equivalent to the shortest-remaining-time-first algorithm

Topic : Multilevel Queue Scheduling

→ use multiple ready Queues

→ add processes in these queues as per some criteria

→ use different algo for different queues.

Topic : Multilevel Queue Scheduling

algo ⇒ RR Q = 2

System Processes →

RR Q = 6

Foreground Processes →

FCFS

Background Processes →

Topic : Multilevel Queue Scheduling

1. Fixed priority preemptive scheduling method

   assign priority to each Queue and execute processes of highest priority queue first.

   once higher priority Queue processes finish then only lower priority Queue processes will run

2. Time slicing

   assign slices to each Queue

Topic : Multilevel Queue Scheduling

fixed priority preemptive

Queue 1: RR with Q=2 (Higher) ~~P1~~, ~~P2~~, ~~P1~~, ~~P2~~, ~~P4~~

Queue 2: FCFS ~~P3~~.

P1	P2	P1	P2	P3	P4	P4	P3

0 2 4 6 7 9 11 13 20

Process	Arrival Time	Burst Time	Queue
P1	0	4	1
P2	0	3	1
P3	0	9	2
P4	9	4	1

Topic : Multilevel Queue Scheduling

fixed priority preemptive

Queue 1: RR with Q=3 ~~P1, P2, P4, P6, P4, P7~~
(Higher)

Queue 2: FCFS P3, P5, P8

P1	P2	P3	P4	P6	P4	P3	P7	P3	P5

0 3 6 10 13 16 17 19 21 23 29

Process	Arrival Time	Burst Time	Queue
P1	0	3	1
P2	0	3	1
P3	2	8	2
P4	10	4	1
P5	11	6	2
P6	11	3	1
P7	19	2	1
P8	13	5	2

$\boxed{P8}$

29 34

Topic : Multilevel Queue Scheduling

Disadvantages:

1. Some processes may starve for CPU if some higher priority queues are never becoming empty

2. It is inflexible in nature. processes can not be moved from one queue to another.

Topic : Multilevel Feedback Queue Scheduling

same as multilevel queue scheduling

but processes can be moved b/w Queues

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used in today's systems

Topic : Multilevel Feedback Queue Scheduling

System Processes

Foreground Processes

Background Processes

Topic : Multilevel Feedback Queue Scheduling

Disadvantage:

1. Some processes may starve for CPU if some higher priority queues are never becoming empty.

Advantage:

1. Flexible

Topic : Analysis of Scheduling Algorithms

Basis of Analysis	Algorithm
Minimum average WT among non-preemptive	SJF
Minimum average WT among all algos	SRTF
Non-preemptive always	FCFS, SJF, LJF, HRRN, Nonpreemptive priority
Preemptive always	None
SRTF behaves as SJF	1. when all processes arrive together 2. when later arriving process is never smaller than current
Preemptive Priority behaves as Non-Preemptive	1. when all processes arrive together 2. when later arriving process never has greater priority.

Topic : Analysis of Scheduling Algorithms

Basis of Analysis	Algorithm
RR behaves as Non-Preemptive	when $Q \geq \max$ (BT of Process)
Convoy Effect	FCFS, LJF
Starvation	SJF, SRTF, LJF, LRTF, Priority based, MLQ, MLFQ

2 mins Summary

Topic Multilevel Queue Scheduling

Topic Multilevel Feedback Queue Scheduling

Happy Learning

THANK - YOU